

Practical-4

Practice lab Assignment

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3.2.5. Numpy: Copying and Viewing Arrays

01:45

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>
View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

copyAnd...

Submit

Debugger

```
1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array =original_array.view()
10
11 # Create a copy
12 copy_array =original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
```

Average time: 0.016 s
Maximum time: 0.020 s

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 20 ms Debug

Expected output: 10 20 30 40 50 60 70 80
Actual output: 10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]
Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]

Terminal Test cases

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4.1.2. Dictionary to dataframe

05:33

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

datafram...

Submit

Debugger

```
1 import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35],
7 }
8
9 # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
```

Average time: 0.149 s
Maximum time: 0.183 s

1 out of 1 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1 183 ms Debug

Expected output: Original DataFrame:
.....Name ..Age
0.....Alice...25
1.....Bob....30
2.....Charlie...35
New name: Susan

Actual output: Original DataFrame:
.....Name ..Age
0.....Alice...25
1.....Bob....30
2.....Charlie...35
New name: Susan

Terminal Test cases

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4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep='\s+', header=None, names=["Name", "Age",
6 "Grade"])
7 print("First five rows:")
8 print(data.head())
9 av=round(data['Age'].mean(),2)
10 print("Average age:",av)
11 fd=data[data['Grade'].isin(['A','B'])]
12 print("Students with a grade up to B")
13 print(fd)
14 # write your code here..
```

Average time0.077 s77.50 msMaximum time0.116 s116.00 ms

1 out of 1 shown test case(s) passed1 out of 1 hidden test case(s) passed

Test case 1116 ms

Expected outputActual output

studentdata.txtstudentdata.txt

First five rows:First five rows:

...Name Age Grade...Name Age Grade

0 John 25 A0 John 25 A

1 Alin 22 B1 Alin 22 B

2 Emma 24 A2 Emma 24 A

TerminalTest cases

< PrevResetSubmitNext>

Lab Assignment

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4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep='\s+', header=None, names=["Name", "Age",
6 "Grade"])
7 print("First five rows:")
8 print(data.head())
9 av=round(data['Age'].mean(),2)
10 print("Average age:",av)
11 fd=data[data['Grade'].isin(['A','B'])]
12 print("Students with a grade up to B")
13 print(fd)
14 # write your code here..
```

Average time0.077 s77.50 msMaximum time0.116 s116.00 ms

1 out of 1 shown test case(s) passed1 out of 1 hidden test case(s) passed

Test case 1116 ms

Expected outputActual output

studentdata.txtstudentdata.txt

First five rows:First five rows:

...Name Age Grade...Name Age Grade

0 John 25 A0 John 25 A

1 Alin 22 B1 Alin 22 B

2 Emma 24 A2 Emma 24 A

TerminalTest cases

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4.2.2. Best Selling Product13:27

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases+

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 ps=df.groupby('Product')['Quantity'].sum()
10
11 best_product = ps.idxmax()
12 highest_quantity = ps.max()
13
14 # Display the result
15 print(f"Best selling product: {best_product}")
```

Average time0.038 s38.33 msMaximum time0.083 s83.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 163 msDebug

Expected output

Actual output

sales_data.csvsales_data.csv

Best selling product: Product ABest selling product: Product A

Total quantity sold: 23Total quantity sold: 23

TerminalTest cases

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4.2.3. City that Sold the Most Products02:20

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases+

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 city=df.groupby('City')['Quantity'].sum()
9
10 # write the code..
11 best_city=city.idxmax()
12 # Display the result
13 print(f"City sold the most products: {best_city}")
```

Average time0.037 s37.00 msMaximum time0.080 s80.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 160 msDebug

Expected output

Actual output

sales_data.csvsales_data.csv

City sold the most products: Los AngelesCity sold the most products: Los Angeles

TerminalTest cases

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4.2.4. Most Frequently Sold Product Pairs

01:50

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appears in transactions on 2025-01-04.

Explorer

frequentli...

```
1 # import pandas as pd
2 # from itertools import combinations
3 # from collections import Counter
4
5 ## Prompt user to input the file name
6 # file_name = input()
7
8 ## Read data from the specified CSV file
9 # df = pd.read_csv(file_name)
10
11 ## write the code
12
13
14 ## Output the most frequent product pairs
15 '''import pandas as pd
```

Average time0.040 s40.00 ms

Maximum time0.085 s85.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 165 msDebug

Expected output

Actual output

sales_data.csvsales_data.csv

Product A and Product B: 2 timesProduct A and Product B: 2 times

Product A and Product C: 2 timesProduct A and Product C: 2 times

TerminalTest cases

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4.2.5. Titanic Dataset Analysis and Data Cleaning

02:30

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `.info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,S
```

Explorer

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # 1. Display the first 5 rows of the dataset
8 print(data.head())
9
10
11 # 2. Display the last 5 rows of the dataset
12 print(data.tail())
13
14
15 # 3. Get the shape of the dataset
```

Average time0.324 s324.00 ms

Maximum time0.324 s324.00 ms

1 out of 1 shown test case(s) passed

Test case 1624 msDebug

Expected output

Actual output

PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
0 1 0 3 Braund, Mr. Owen Harris male 22 1 0 A/5 21171 7.25 S
1 NaN NaN 5
2 1 2 1 Cumings, Mrs. John Bradley (Florence Briggs Thayer) female 38 1 0 PC 17599 71.2833 C85 C
3 NaN NaN 5

PassengerId Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked
0 1 0 3 Braund, Mr. Owen Harris male 22 1 0 A/5 21171 7.25 S
1 NaN NaN 5
2 1 2 1 Cumings, Mrs. John Bradley (Florence Briggs Thayer) female 38 1 0 PC 17599 71.2833 C85 C
3 NaN NaN 5

TerminalTest cases

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
```

Explorer

titanicDat...

Submit

Debugger

```
1 # import pandas as pd
2 # import numpy as np
3
4 ## Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 ...
8 import pandas as pd
9 import numpy as np
10
11 # Load the Titanic dataset
12 data = pd.read_csv('Titanic-Dataset.csv')
13 data['FamilySize'] = data['SibSp'] + data['Parch']
14
15 ...
```

Average time0.158 sMaximum time0.158 s1 out of 1 shown test case(s) passed

Test case 1158 msDebug

Expected outputActual output

29.6991176470588229.69911764705882

14.454214.4542

3...4913...491

1...2161...216

2...1842...184

Name: Pclass, dtype: int64Name: Pclass, dtype: int64

TerminalTest cases

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarkation location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
```

Explorer

titanicDat...

Submit

Debugger

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10 # 1. Calculate the survival rate by class
11 print(data.groupby('Pclass')['Survived'].mean())
12
13
14 # 2. Calculate the survival rate by embarked location
15 print(data.groupby('Embarked_S')['Survived'].mean())
```

Average time0.162 sMaximum time0.162 s1 out of 1 shown test case(s) passed

Test case 1162 msDebug

Expected outputActual output

PclassPclass

1...0.6296301...0.629630

2...0.4728262...0.472826

3...0.2423633...0.242363

Name: Survived, dtype: float64Name: Survived, dtype: float64

Embarked_SEmbarked_S

TerminalTest cases

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 401/12

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
```

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8 import pandas as pd
9 import numpy as np
10
11 # Load the Titanic dataset
12 data = pd.read_csv('Titanic-Dataset.csv')
13 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
14
15 import pandas as pd
16 import numpy as np
```

Average time0.162 s162.00 ms

Maximum time0.162 s162.00 ms

1 out of 1 shown test case(s) passed

Test case 1162 msDebug

Expected output

Actual output

female----233

female----233

male-----109

male-----109

Name: Sex, dtype: int64

Name: Sex, dtype: int64

male-----468

male-----468

female-----81

female-----81

Name: Sex, dtype: int64

Name: Sex, dtype: int64

TerminalTest cases

< PrevResetSubmitNext >